

Activity 8 - HackerRank Algorithmic Problem-Solving & Portfolio

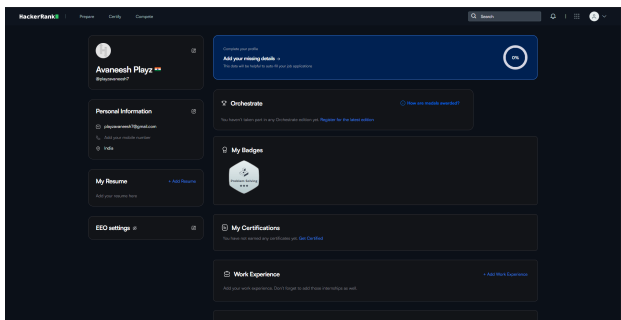
3rd Semester B.Tech CSE • Studio Deliverable

Student Name	Avaneesh K Rao	Student ID	Not provided	Section	Not provided
GitHub Repository	Not provided				
HackerRank Profile	https://www.hackerrank.com/profile/playzavaneesh7				
Achievement	3-Star Problem Solving badge (verified on profile screenshot)				

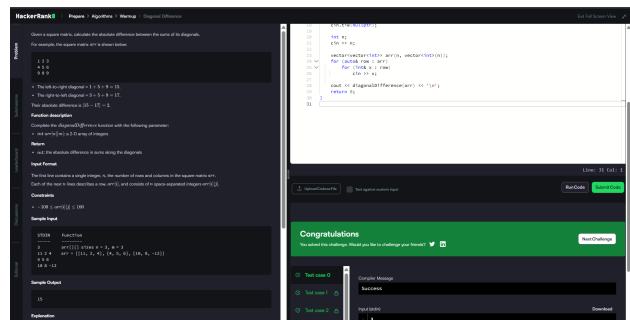
Mandatory Problems & Complexity

Problem	Topic	Time	Space
Diagonal Difference	2D Arrays / Matrices	$O(N)$	$O(1)$
Dynamic Array	Data Structures / Vectors	$O(N + Q)$	$O(N)$
Time Conversion	Strings & Logic	$O(1)$	$O(1)$
Compare the Triplets	Basic Implementation	$O(1)$	$O(1)$
Sparse Arrays	Hash Maps / Strings	$O(N + Q)$	$O(N)$

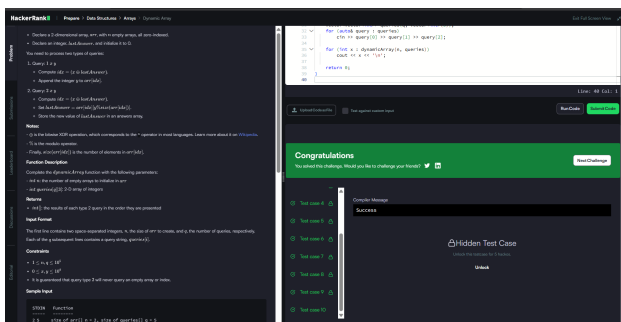
HackerRank Evidence & Badge



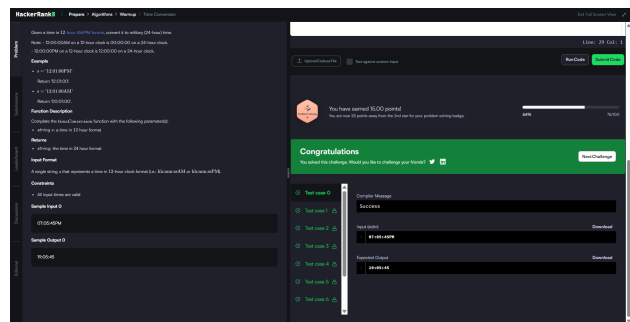
HackerRank profile showing the 3-Star Problem Solving badge



Accepted: Diagonal Difference (all test cases passed)



Accepted: Dynamic Array (all test cases passed)

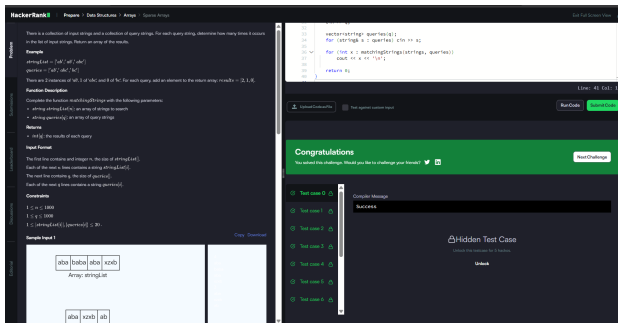


Accepted: Time Conversion (all test cases passed)

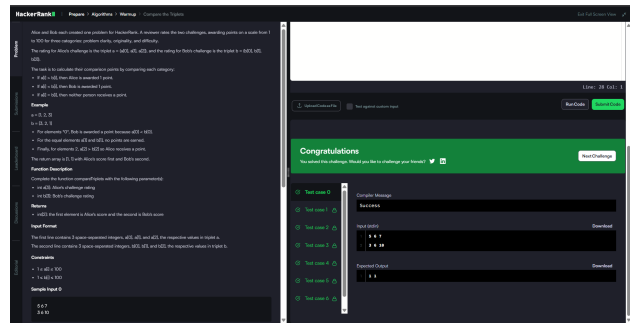
Technical Documentation & Reflection

HackerRank evidence, solution organization, and algorithmic optimization reflection

Additional HackerRank Evidence



Accepted: Sparse Arrays (all test cases passed)



Accepted: Compare the Triplets (all test cases passed)

GitHub Repository Structure

The portfolio is intended to contain separate folders for the five mandatory HackerRank problems, source-code files, accepted-submission screenshots, badge evidence, and a README documenting the work. The GitHub link was not visible in the supplied screenshots, so it has not been invented in this report.

Reflection on Algorithmic Optimization

In this activity I practiced solving HackerRank problems by focusing on correctness, efficiency, and input size. Diagonal Difference and Compare the Triplets use direct, constant-work comparisons, while Dynamic Array and Sparse Arrays demonstrate how suitable data structures reduce unnecessary repeated work. I also practiced string processing in Time Conversion and permutation logic in additional HackerRank problems. Testing against HackerRank's visible and hidden cases helped verify that the implementations were both correct and efficient. Preparing separate source files and collecting accepted-submission evidence also made the solutions easier to organize and present as a portfolio.

Learning Outcomes

- Analyzed time and space complexity of programming solutions.
- Practiced arrays, vectors, strings, matrices, and query-based problems.
- Used suitable data structures to avoid unnecessary operations.
- Verified solutions using HackerRank test cases.
- Organized source code and evidence for portfolio presentation.

Submission Links

GitHub Repository	Not provided in supplied screenshots
HackerRank Profile	https://www.hackerrank.com/profile/playzavaneesh7